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Advancing body composition assessment in patients with cancer: First comparisons of traditional versus multicompartment models

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ABSTRACT

Background and aims: Measurement of body composition using computed tomography (CT) scans may be a viable clinical tool for low muscle mass assessment in oncology. However, longitudinal assessments are often infeasible with CT. Clinically accessible body composition technologies can be used to track changes in fat-free mass (FFM) or muscle, though their accuracy may be impacted by cancer-related physiological changes. The purpose of this study was to examine the agreement among accessible body composition method with criterion methods for measures of whole-body FFM measurements and, when possible, muscle mass for the classification of low muscle in patients with cancer.

Methods: Patients with colorectal cancer were recruited to complete measures of whole-body DXA, air displacement plethysmography (ADP), and bioelectrical impedance analysis (BIA). These measures were used alone, or in combination to construct the criterion multicompartment (4C) mode for estimating FFM. Patients also underwent abdominal CT scans as part of routine clinical assessment. Agreement of each method with 4C model was analyzed using mean constant error (CE = criterion − alternative), linear regression including root mean square error (RMSE), Bland-Altman limits of agreement (LoA) and mean percentage difference (MPD). Additionally, appendicular lean soft tissue index (ALSTI) measured by DXA and predicted by CT were compared for the absolute agreement, while the ALSTI values and skeletal muscle index by CT were assessed for agreement on the classification of low muscle mass.

Results: Forty-five patients received all measures for the 4C model and 25 had measures within proximity of clinical CT measures. Compared to 4C, DXA outperformed ADP and BIA by showing the strongest overall agreement (CE = 1.96 kg, RMSE = 2.45 kg, MPD = 98.15 ± 2.38%), supporting its use for body composition assessment in patients with cancer. However, CT cutoffs for skeletal muscle index or CT-estimated ALSTI were lower than DXA ALSTI (average 1.0 ± 1.2 kg/m^2) with 24.0% to 32.0% of patients having a different low muscle classification by CT when compared to DXA.

Conclusions: Despite discrepancies between clinical body composition assessment and the criterion multicompartment model, DXA demonstrates the strongest agreement with 4C. Disagreement between DXA and CT for low muscle mass classification prompts further evaluation of the measures and cutoffs used with each

Abbreviations: %FM, percent fat mass; 4C, 4-compartment model; ADP, air displacement plethysmography; ALST, appendicular lean soft tissue; ALSTI, appendicular lean soft tissue index; BIA, bioelectrical impedance analysis; BM, body mass; BMC, bone mineral content; BV, body volume; CCC, concordance correlation coefficient; CE, constant error; CT, computed tomography; CV, coefficient of variation; DXA, dual-energy X-ray absorptiometry; FFM, fat-free mass; FFM_{hyd}, hydration of fat-free mass; FM, fat mass; Ht, height; LoA, limits of agreement; M, mineral; MPD, mean percentage difference; R, resistance; RMSE, root mean square error; SD, standard deviation; SMI, skeletal muscle index; TBW, total body water; Xc, reactance

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technique. Multicompartment models may enhance our understanding of body composition variations at the individual patient level and improve the applicability of clinically accessible technologies for classification and monitoring change over time.

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Introduction

Evaluating body composition in patients with cancer is crucial, especially when diagnosing low muscle mass and its related conditions. The often overlooked presence of low muscle mass in these patients, even within the context of cancer treatments, has severe implications [1]. Low muscle mass is a phenotypic criterion for malnutrition [2] and both conditions are associated with mortality, highlighting the importance of comprehensive nutritional assessment in oncology and other clinical settings [3–5]. In the context of cancer care, early identification of low muscle mass helps inform best practices for risk assessment and optimized treatment strategies [6]. Some studies using accessible measures, including body mass index (BMI), have reported a protective effect of increased BMI on survival. However, a high BMI mainly relates to the fat mass (FM) compartment and a low BMI is generally a reflection of the muscle mass compartment. As such, these protective effects are negated after controlling for body composition [7–9]. In addition, decreased mobility is prevalent even among patients with obesity, due to an imbalance of FM and fat-free mass (FFM), and is linked to compromised functional status and survival [10]. These findings highlight not only the importance of body composition assessment for risk stratification at diagnosis but also for monitoring disease-related malnutrition during treatment and/or disease progression [11].

Computed tomography (CT) is commonly used for evaluating the prevalence of low muscle mass in the trunk (third lumbar vertebrae) at the time of cancer diagnosis in research settings and has been increasingly used in clinical settings [10,12]. However, its application for ongoing monitoring of body composition is limited due to the availability of these images in patient's medical records, as well as limitations related to prospective evaluations, such as radiation exposure and cost. Consequently, use of CT for this purpose is typically limited to occasions when it is already being employed in cancer staging/monitoring [12,13]. Commonly used methods that can be used for the measurement of FFM or muscle mass include dual-energy X-ray absorptiometry (DXA) and bioelectrical impedance (BIA). Notably, these are recommended methods for the assessment of the low muscle mass phenotypic criterion within the context of the Global Leadership Initiative on Malnutrition diagnosis [14,15]. Despite being less commonly employed in oncology, air displacement plethysmography (ADP) has been an effective method for body composition quantification [16,17]. However, these clinical methods involve two-compartment (2C) estimations of FFM that assume that each of the FFM components (the non-fat components of bone, skeletal muscle, organs, skin, etc.) is in constant proportion. FFM using these techniques relies on an average hydration (~73.2%) of the FFM compartment (FFM_{hyd}) [18]. FFM_{hyd} is known to vary based on FFM composition, race, age, nutritional status, and disease state, potentially confounding single timepoint estimations of FFM from the 2C approach [19–23]. Additionally, changes in FFM_{hyd} may make 2C approaches imprecise for longitudinal measures across a range of pathological conditions [24,25].

To overcome limitations of traditional 2C assessment methods, multicompartment body composition models offer enhanced accuracy in measurements and the capability to calculate FFM_{hyd} [26]. FFM_{hyd} is based on the composition of total body water (TBW), bone mineral, and protein, thus reflecting nutritional and overall health status. A deviation in the typical FFM_{hyd} can indicate malnutrition or other underlying health conditions commonly found in cancer. Moreover, FFM_{hyd} is essential for a precise differentiation between FFM and FM. In clinical settings, changes in FFM_{hyd} can also act as markers of disease progression or the efficacy of certain treatments [27]. As such, it is crucial both for research and clinical practice to accurately understand how 2C models perform in populations with expected variations in FFM_{hyd}. This understanding is vital for developing more sophisticated models that can more effectively monitor changes in body composition in clinical settings. Accurate monitoring is key for tailoring patient care, understanding the progression of conditions, and evaluating the efficacy of interventions. This in turn enhances the quality of patient management and treatment outcomes, as well as facilitates the understanding of effective intervention strategies. To date, no studies have employed a multicompartment model to evaluate traditional 2C assessment methods for FFM accuracy in patients with cancer. In this study, we evaluated performance of three body composition techniques against a multicompartment model. Additionally, we compared differences in low muscle mass classification using muscle mass values estimated from CT and DXA measures.

Methods

We used baseline data from the Protein Recommendations to Increase Muscle (PRIMe) randomized controlled pilot trial (clinicaltrials.gov NCT02788955) exploring the feasibility and effects of diets with different level of protein (2 g/kg/d versus 1 g/kg/d protein intake) on muscle mass maintenance during 12 wk of chemotherapy [28]. A detailed trial protocol is published elsewhere [29]. A total of 50 patients with stage II-IV colorectal cancer without cachexia, capable of oral intake, and undergoing anti-cancer therapy were recruited at the Cross Cancer Institute (Alberta, Canada). The trial was approved by the Health Research Ethics Board of Alberta-Cancer Committee (HREBA.CC-15-0193) and complied with standards on the use of human participants in research. All patients provided written informed consent.

Prior to trial randomization and intervention, patients completed baseline clinical and body composition assessments at the Human Nutrition Research Unit (University of Alberta, Canada). Pretest protocols required fasting for 10 h, avoiding alcohol, and minimizing physical activity for 24 h prior to assessments. Height was measured on a 235 Heightronic Digital Stadiometer (Quick Medical, Issaquah, Washington, USA) to the nearest 0.1 cm. Body weight measured in triplicate to the nearest 0.1 kg on a calibrated digital scale (Health-o-meter Professional Remote Display; Sunbeam Products Inc., Florida, USA) with patients wearing thin, light clothing was averaged. Electronic medical records provided patient characteristics including age, sex, and disease stage. Ethnicity was self-reported using a study-specific questionnaire.

All patients who completed baseline trial assessments are included in this analysis. The purpose of this analysis was to compare baseline FFM measurements to understand agreement of 2C methods to the criterion (4C). This was assessed through group-level agreement, individual error, and the relative agreement to

the criterion. Individual agreement and overall relative error were employed to demonstrate the 2C method that exhibited the strongest absolute agreement with the criterion measure. Additionally, because DXA provides appendicular lean soft tissue (ALST) measures, recommended for identifying low muscle mass in clinical practice, measures of muscle mass by CT and DXA were compared for classification of low muscle mass and to assess DXA's potential suitability as a longitudinal clinical assessment tool in place of CT.

Measurements collected

Air displacement plethysmography

Patients completed ADP assessment using a BOD POD Gold Standard Body Composition Tracking System (COSMED USA, Inc., Concord, California, USA). Patients wore thin, tight clothing and a swim cap to minimize air on the body surface and in materials (e.g., clothing). Manufacturer standard protocols were followed for pretest quality control assessment and actual assessment. Thoracic gas volume was predicted for all patients using device software.

ADP provides criterion body volume (BV) measures, establishing it as a benchmark for BV assessment in multicompartment body composition modeling Additionally, ADP is also considered a highly accurate tool for FFM estimation [30,31]. Device software calculated BV based on measured mass and density. The software calculated fat mass percent (%FM) using the Siri equation and FFM was subsequently derived using FM and body weight [32].

Dual-energy X-ray absorptiometry

Patients completed a DXA scan using a General Electric Lunar iDXA High Speed Digital Fan Beam Densitometer with Encore 13.60 software (General Electric Company, Madison, Wisconsin, USA). DXA systems quantify bone mineral content (BMC) and the remaining body mass as soft tissue, making it fundamentally a 2C model. That said, DXA is often considered a three-compartment model due to the ability to estimate lean soft tissue (LST) and FM. Each scan provides BMC, LST, and FM at whole-body and regional levels. FFM obtained from DXA is the sum of LST and BMC. ALST represents the sum of bone-free LST from the limbs. ALST was standardized by height^2 to provide ALSTI. Whole-body BMC was utilized in the 4C model, as further outlined below.

Bioelectrical impedance analysis

Due to the availability of devices and changes in Canadian regulations regarding use of medical technologies, and devices not being updated to align with these regulations, three different BIA devices were used for separate subsets of patient data collection. All three devices were phase sensitive: two supine tetrapolar BIA (Vitality Analyzer [Nutritional Solutions Corporation, Ltd; Harrisville, MI, USA using GE MACTRODE 9623-003P electrocardiogram electrodes; n = 11], Bodystat Quadscan 4000 [Bodystat Ltd, Isle of Man, United Kingdom using Bodystat 104-0525 Quadscan electrodes; n = 13]) and a supine octopolar (Seca mBCA 525 [Seca GmBH, Hamburg, Germany using Kendall/Covidien Medi-tract Mini 130 electrocardiogram electrodes; n = 21]). Raw impedance measures of resistance (R) and reactance (Xc) were recorded at 50 kHz for the right side of the body or averaged for the left and right side (Seca device).

To avoid errors associated with the use of separate manufacturer-specific proprietary equations, we identified TBW prediction equations based on the resistance index (height^2 divided by resistance; Ht^2/R) at 50 kHz. Only two studies have developed empirical equations for TBW prediction using BIA in patients with cancer [33,34]. Additionally, Fredrix et al. evaluated multiple published equations and concluded that the equation by Lukaski et al. provided the most reasonable estimates for TBW in these patients [34,35]. These equations are provided in Supplemental Table 1. Wells et al. [36] showed that the aggregate (average) of multiple TBW prediction equations randomly distributes error across predictions. Because errors tend to cancel out, they showed that the aggregate approach increased the accuracy of TBW predictions. Therefore, we used an average of the three prediction equations to calculate TBW in the multicompartment model. For estimates of FFM using BIA, we assumed constant hydration FFM_{hyd} = 73.2%, calculating FFM = TBW/0.732.

Computed tomography

CT measurements taken at the time of cancer diagnosis during routine clinical assessments were retrospectively obtained from patients' medical records. Measurements were performed at the L3 vertebra and included skeletal muscle cross-sectional area (SM_{CT}) and muscle radiodensity (indicative of muscle composition) of the following combined muscle groups: rectus and transversus abdominus, internal and external obliques, psoas major and minor, quadratus lumborum and erector spinae. Cross-sectional area of the visceral adipose, subcutaneous adipose, and intramuscular adipose tissue groups was also quantified. Muscle and adipose tissues were evaluated based on differences in pre-established measures of Hounsfield unit attenuations as follows: −29 to +150 for skeletal muscle, −150 to −50 for visceral adipose tissue, and −190 to −30 for subcutaneous and intermuscular adipose tissues [37,38]. Images of the L3 vertebra were landmarked using sliceOmatic (Tomovision, Magog, Canada, version 5.0) as described in detail elsewhere [39]. With the later availability of Data Analysis Facilitation Suite (DAFS; Voronoi Health Analytics, Vancouver, Canada, version 3.8), automated tissue segmentation was performed using the above-mentioned Hounsfield units and manual editing was applied, as needed.

Skeletal muscle index (SMI) from spinus muscle CT was calculated as skeletal muscle area (in cm^2) divided by height^2. Cut points for low SMI in cancer patients have previously been published [8]. Though SMI is used in research for identifying low muscle mass, CT has also been used to predict regional muscle mass, with low ALST being used as part of malnutrition or sarcopenia assessments [40]. A previously published equation by Mourtzakis et al. [41] was used to indirectly calculate ALST (termed appendicular skeletal muscle in the Mourtzakis paper), which can be used for direct comparison to DXA ALSTI. The calculation for height-standardized ALST, (ALSTI) is:

$$ALSTI_{CT}(kg/m^2) = 0.11 * SM_{CT}/height^2 + 1.17; r = 0.89, SEE = 0.10 kg/m^2$$

FFM (from the Mourtzakis paper) were also calculated for direct comparisons to DXA and 4C using the following equation:

$$FFM_{CT}(kg/m^2) = 0.14 * FFM_{CT} + 0.72; r = 0.83, SEE = 1.2 kg$$

CT measurements performed less than 90 d from DXA measures were compared to maintain proximity between measures.

Multicompartment body composition

The multicompartment model of body composition serves as the criterion FFM measurement. Multicompartment models

include measures of TBW to adjust for FFM_{hyd} as the density of FFM differs due to variation in the three FFM compartments of water, protein, and bone mineral (with FM being the fourth compartment) [42]. The Wang et al. 4C model [42] utilizes criterion deuterium dilution (D_2O) measures of TBW, BMC from DXA, BV from ADP, and body mass (BM) from scale weight to obtain FM. Because the criterion method for TBW assessment, deuterium dilution (D_2O), was not collected in this study, we utilized BIA predicted TBW as it has been shown to be a valid alternative approach for use in multicompartment modeling [43]. Mineral (M) is calculated by multiplying DXA BMC by 1.0436 [44]. The 4C model for FM was calculated using the following equation [42]:

$$FM_{4C} = 2.748 * BV - 0.699 * TBW + 1.129 * M - 2.051 * BM$$

FFM_{4C} was derived by subtracting the FM_{4C} from body mass and FFM_{hyd} was calculated by dividing TBW by FFM_{4C}.

All measures for the multicompartment model were performed within close proximity to ensure robust data quality. ADP and BIA measurements were performed on the same day, while DXA measurements were performed, on average, within 2.4 ± 3.1 d.

Statistical methods

To assess differences between the TBW estimated from each of the three BIA equations, we assessed mean differences using within group one-way ANOVA with Bonferroni correction ($P < 0.05/3 = 0.0167$). After the calculation of TBW, the BV, M, and BM were entered into the 4C equation to calculate FM. FFM was calculated as weight − FM.

With 4C as the criterion, estimation of FFM of each clinical 2C method was compared using paired samples t-tests and constant error (CE = actual FFM − predicted FFM) for group-level agreement. Additionally, linear regression (slope, intercept, coefficient of determination [r^2], root mean square error [RMSE]) and Lin's Concordance Correlation Coefficient (CCC) were performed. The range of individual agreement was assessed by constructing Bland-Altman comparisons with 95% limits of agreement (LoA). To compare the similarity of results to the criterion 4C, we calculated the mean percentage difference (MPD = [((criterion + estimate)/2)/criterion] * 100. Calculation of the standard deviation (SD) of the MPD allows for the calculation of the coefficient of variation (%CV = SD/MPD), allowing for the direct comparison of multiple methods to the criterion [45]. An MPD nearer to 100 and minimal %CV indicate greater agreement to the criterion. The LoA, RMSE, and MPD were evaluated to identify the 2C method with the strongest overall agreement to the criterion.

CT prediction estimates of ALSTI were compared to those from DXA (due to DXA's measurement of lean soft tissue and subsequent recommendations for low muscle mass assessment) to determine agreement of absolute measures. Further, the prevalence of low muscle mass of CT SM (<52.3 cm^2/m^2 (m) or <38.6 cm^2/m^2 (f) if BMI <30 kg/m^2, or <54.3 cm^2/m^2 (m), <46.6 cm^2/m^2 (f) if BMI ≥30 kg/m^2) and CT-predicted ALSTI were compared to those from DXA using with widely used cut points (ALSTI <7.0 kg/m^2 in males and <5.5 kg/m^2 in females) [8,46]. A kappa coefficient was calculated to evaluate the agreement between the CT and DXA low muscle classifications. All analyses were performed with SAS Version 9.4 (SAS Institute, Cary, IN).

Results

Of this sample, two did not perform ADP analysis, and three did not receive BIA measurements. Removing these individuals, a total of 45 patients aged 57.7 ± 11.0 y (9 patients ≥65 y), height 171.1 ± 9.2 cm, weight 79.2 ± 17.1 kg, BMI 27.1 ± 5.5 kg/m^2 had available data for use in the multicompartment model analyses (Table 1). Patients were classified by BMI in the underweight (n = 1, 2.2%), normal weight (n = 15, 33.3%), overweight (n = 16, 35.6%), or obesity (n = 13, 28.9%) categories. Patients were diagnosed with stage 2 (n = 5, 11.1%), stage 3 (n = 28, 62.2%), or stage 4 (n = 12, 26.7%) colorectal cancer. The range of FFM_{hyd} was 62.7 to 79.6%, with a significant mean difference between sexes (P = 0.02). Of 45 patients, 43 (95.6%) had a FFM_{hyd} less than the reported average of 73.2%.

Prior to constructing the 4C model, agreement between the three BIA prediction equations for TBW was evaluated (Supplemental Fig. 1). Though the three equations did not differ significantly ($p > 0.0167$), the Simons equation (38.2 ± 7.0 L) consistently predicted a larger volume of TBW than the Fredrix (36.6 ± 6.8 L) or Lukaski (36.9 ± 8.6 L) equations. Averaging values obtained from the three equations removed the disagreement between equations at low and high TBW values.

Comparisons of FFM to the criterion 4C model are presented in Table 2 and Bland-Altman comparisons to the 4C model are presented in Figure 1. While ADP showed a better mean agreement (CE = 0.34, P = 0.16), this method exhibited a significant bias (average overestimation of FFM > 60 kg) compared to 4C. DXA and BIA had greater CE (1.96 kg for DXA, 2.97 kg for BIA, $P < 0.01$ for both). BIA also showed a significant (P = 0.02) bias, with underestimations at greater FFM. Overall, DXA showed the lowest RMSE

Table 1
Demographics of 45 patients with colorectal cancer and multicompartment model data (n = 45)

	Total (n = 45) Mean ± SD	Males (n = 27) Mean ± SD	Females (n = 18) Mean ± SD	*P* difference between sexes
Age (y)	57.7 ± 11.0	59.9 ± 9.8	54.3 ± 12.0	0.11
Height (cm)	171.1 ± 9.2	176.3 ± 7.0	163.1 ± 5.7	**<0.001**
Weight (kg)	79.2 ± 17.1	83.8 ± 17.7	72.4 ± 14.0	**0.02**
BMI (kg/m^2)	27.1 ± 5.5	26.9 ± 5.2	27.4 ± 6.2	0.80
Ht^2/R (cm^2/Ω)	55.4 ± 13.7	63.4 ± 11.3	43.3 ± 5.8	**<0.001**
FFM_{4C} (kg)	53.9 ± 11.5	60.9 ± 9.3	43.6 ± 4.4	**<0.001**
FM_{4C} (kg)	26.0 ± 10.7	24.1 ± 10.0	28.7 ± 11.3	0.17
$\%FM_{4C}$	31.9 ± 9.8	27.4 ± 7.9	38.3 ± 8.8	**<0.001**
$FFM_{hyd\text{-}4C}$ (%)	69.4 ± 3.3	68.8 ± 3.5	70.2 ± 2.9	**0.02**
$ALSTI_{DXA}$ (kg/m^2)	7.4 ± 1.5	8.1 ± 1.4	6.5 ± 0.9	**<0.001**

%FM, percent fat mass; 4C, 4-compartment model; ALSTI, appendicular lean soft tissue index; BIA, bioelectrical impedance analysis; FFM, fat-free mass; FFM_{hyd}, hydration of fat-free mass; FM, fat mass; Ht, height; NH, Non-Hispanic; R, resistance.
Significant differences were assessed using independent samples *t* test, with significant values ($p < 0.05$) bolded.
Ethnicities: NH Black (n = 1), NH White (n = 35), Filipino (n = 1), Latin American (n = 2), Indigenous Peoples of Canada (n = 4), South Asian (n = 1), and Other (n = 1).

Table 2
Comparisons of body composition estimates of fat-free mass to the criterion multicompartment model (n = 45)

	FFM estimates (kg)					Agreement					
Method	Mean ± SD	Slope	Intercept	r^2	RMSE	CCC	CE	Lower Limits	Upper Limits	MPD ± SD, %	%CV
4C	53.86 ± 11.49										
$2C_{ADP}$	53.52 ± 11.94	0.94	3.62	0.95	2.58	0.96	0.34	−4.97	5.66	100.99 ± 5.37	5.32
$2C_{DXA}$	51.91 ± 11.67*	0.96	3.86	0.96	2.45	0.97	1.96	−2.98	6.87	98.15 ± 2.34	2.38
$2C_{BIA}$	50.89 ± 10.36*	1.08	-1.13	0.94	2.65	0.93	2.97	−2.53	8.48	97.38 ± 2.77	2.83

*Significant differences ($P < 0.05$) were assessed using paired samples *t* test.

(DXA = 2.45 kg, ADP = 2.58 kg, BIA = 2.65 kg) and smaller LoA (range: DXA = ± 4.92 kg, ADP = ± 5.32 kg, BIA = ± 5.50 kg) compared to ADP or BIA. DXA also showed the greatest agreement to 4C using MPD and %CV. The total number of patients who were within 1 kg of the 4C FFM were: DXA (n = 15, 33.3%), ADP (n = 13, 28.9%), and BIA (n = 13, 28.9%).

Further examination of hydration effects was explored by graphing the comparison of FFM_{hyd} to difference in FFM (4C) in Supplemental Figure 2. Both ADP and DXA showed high variability in agreement to 4C across FFM_{hyd}, though variability due to hydration was less pronounced by ADP (r = −0.10) compared to DXA (r = −0.26). For DXA, all but 7 of the 43 individuals with FFM_{hyd} less than 73.2% had FFM underestimated compared to 4C. Because we applied the assumed constant of 73.2% for BIA, it exhibited a strong relationship of FFM_{hyd} variance and FFM agreement (r = −0.96, $P < 0.01$).

In the dataset, 25 patients had CT and DXA conducted within 90 d (average of 60.1 ± 21.3 d). Supplemental Table 2 shows that use of CT equations to estimate FFM resulted in underestimation of FFM (−5.9 ± 6.8 kg) and ALST (−3.2 ± 3.7 kg) compared to 4C and DXA, respectively. Figure 2 shows the difference in ALSTI measured by DXA versus predicted by CT. Overall, CT prediction significantly ($P < 0.01$) underestimated ALSTI by 1.0 ± 1.2 kg/m^2, with the degree of error increasing as ALSTI increased (significant bias, $P < 0.01$, Fig. 2b). The box-whisker plots showed that the majority (20 of 25, 80%) of patients had a lower ALSTI estimate from CT, with underestimation variability extending up to 3.4 kg/m^2. For those categorized as having low muscle mass by CT but not by DXA, ALSTI differences were up to 2.4 kg/m^2 higher as assessed by DXA.

Using published thresholds for low muscle mass, 14 (56.0%) patients were below low muscle thresholds for CT, with 2 fewer females below DXA thresholds when using CT-predicted ALSTI (n = 12, 48.0%). Use of DXA resulted in 8 (32.5%) patients below published thresholds (Table 3).

Error matrices of the agreement between methods are presented in Table 4. In the CT- and DXA-derived low muscle mass comparison, of those with low DXA muscle mass, seven of the eight were also deemed as having low muscle mass by CT. Using DXA, all eight patients were in the underweight (n = 1) or normal weight (n = 7) BMI categories. However, in patients only deemed as having low muscle by CT, five of the seven were in the overweight (n = 3) or obesity (n = 2) BMI categories. The kappa score for agreement between CT and DXA, showing agreement in 17 of 25 patients, was 0.39, which was greater in the comparisons for ALSTI between methods (kappa = 0.51) due to the two fewer instances of patients being classified as having low muscle by CT when DXA assessments suggested normal muscle levels.

Discussion

Limited research has examined relationships among body composition assessment methods in patients with cancer, with most studies using DXA or ADP as reference methods [47–50], this is the first study to employ multicompartment modeling, a criterion approach to *in vivo* body composition quantification. This study examined agreement between clinically-accessible 2C methods (ADP, BIA, DXA) to the criterion 4C model. We show that higher absolute agreement and smaller overall individual error by DXA provides the most accurate estimation of FFM. Though DXA provides a more accurate and accessible body composition model for use in clinical practice, we observed discrepancies between CT assessments of spinal muscle area (and mass) and DXA regional measurements, both commonly used to identify low muscle mass and monitor muscle loss.

In patients with cancer, accurate body composition assessment is critical for monitoring disease-related metabolic changes such as muscle loss (including in the context of malnutrition or cachexia) and for guiding nutritional support [51]. Muscle mass is the primary component of LST and FFM, which is the main site for chemotherapeutic drug metabolism and associated toxicity [52]. Moreover, the amount of muscle is also related to prognosis in various cancer types [53,54]. Therefore, selecting the proper method for FFM assessment and monitoring of change is essential in oncology populations [55]. While the use of cross-sectional CT images of the L3 vertebrae to predict muscle mass are considered a valid and accurate method to assess body composition, its use for longitudinal assessment is extremely limited due to availability/timing, cost, and radiation exposure [12]. This underscores the need for alternative methods available during routine clinical assessment, enabling precise monitoring of body composition alterations due to disease progression, as well as the impact of treatments/ interventions, including those related to nutrition and physical activity. Thus, more work is necessary to understand these differences before DXA and CT can be considered interchangeable for longitudinal assessments of body composition, particularly low muscle mass.

Findings of this study highlight the disagreement between CT and DXA for body composition estimation when L3 measures from CT are used to predict whole-body (or regional) composition. Tewari reported a significantly greater FFM by CT in 50% of their healthy subjects [56]. Kilgour et al. [57] reported that 53% of L3-based CT predictions of ALST were lower than DXA, with an average underestimation of −0.4 ± 1.0 kg in patients with advanced cancers. Conversely, our study found that 20 patients (80%) were underestimated using CT-predicted ALST, with the average underestimation to DXA of 3.2 ± 3.7 kg. These differences have significant implications when applying results to commonly referenced cut offs for low muscle mass. For example, in a sample of 120 Asian patients with gastric cancer, Kim et al. [58] observed differences in low muscle mass prevalence between CT skeletal muscle (63.6%) and DXA (based on the use of European Working Group on Sarcopenia [43.3%] or Asian Working Group on Sarcopenia [40.8%] recommended cut offs). Their study showed a strong correlation between DXA and CT-predicted ALST, though with larger

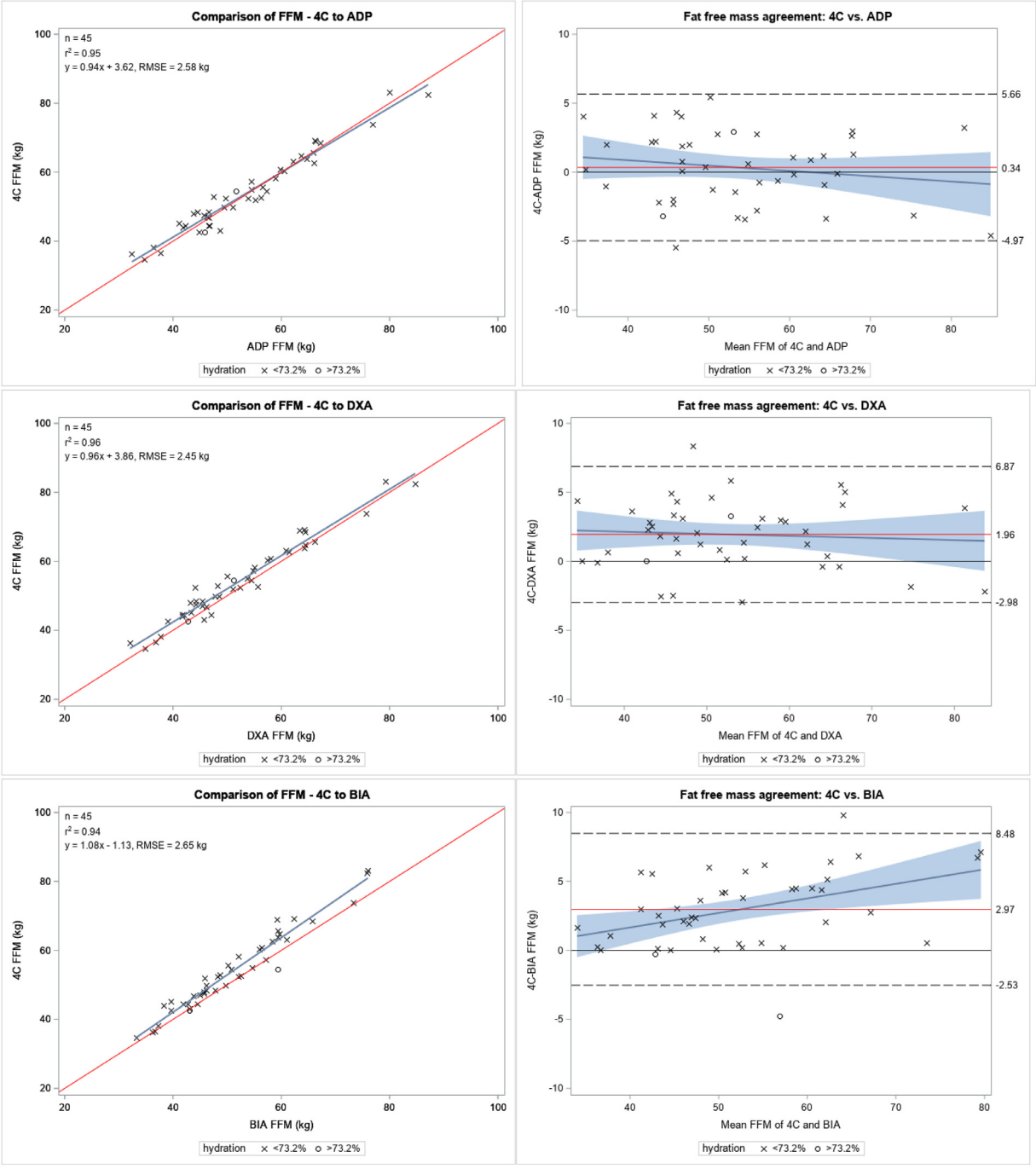

Fig. 1. Comparison of fat-free mass between criterion (4C) and 2C models (n = 45). 4C, 4-compartment; ADP, air displacement plethysmography; BIA, bioelectrical impedance analysis; DXA, dual-energy X-ray absorptiometry; FFM, fat-free mass; RMSE, root mean square error.

individual differences (exact errors were not reported). Our study adds to this work by showing the absolute difference in ALSTI by CT-prediction and DXA within subjects as well as how those differences impact the interpretation of nutritional status and risk of low muscle mass in this patient population. These findings further prompt the need for greater understanding of factors that impact the agreement between body composition modalities and/or the need for more accurate predictive equations.

Though we show that whole-body FFM by DXA (53.1 ± 13.9 kg) presents group-level inaccuracies when compared to 4C FFM (54.8 ± 13.5 kg), the average error was much closer than that as predicted by CT (49.0 ± 12.5 kg). While studies have validated the correlation between DXA-derived ALST and SM from CT or MRI (r = 0.77–0.95, $P < 0.001$) [59,60], differences between methods may result from the impact of hydration on both CT and DXA estimations of lean components. Abnormal hydration levels, common in patients with cancer due to losses of protein, mineral, or water, and/or from edema-related overhydration, are known to impact FFM_{hyd} [61]. However, comparisons of DXA with the 4C model have consistently indicated that discrepancies in FFM estimation

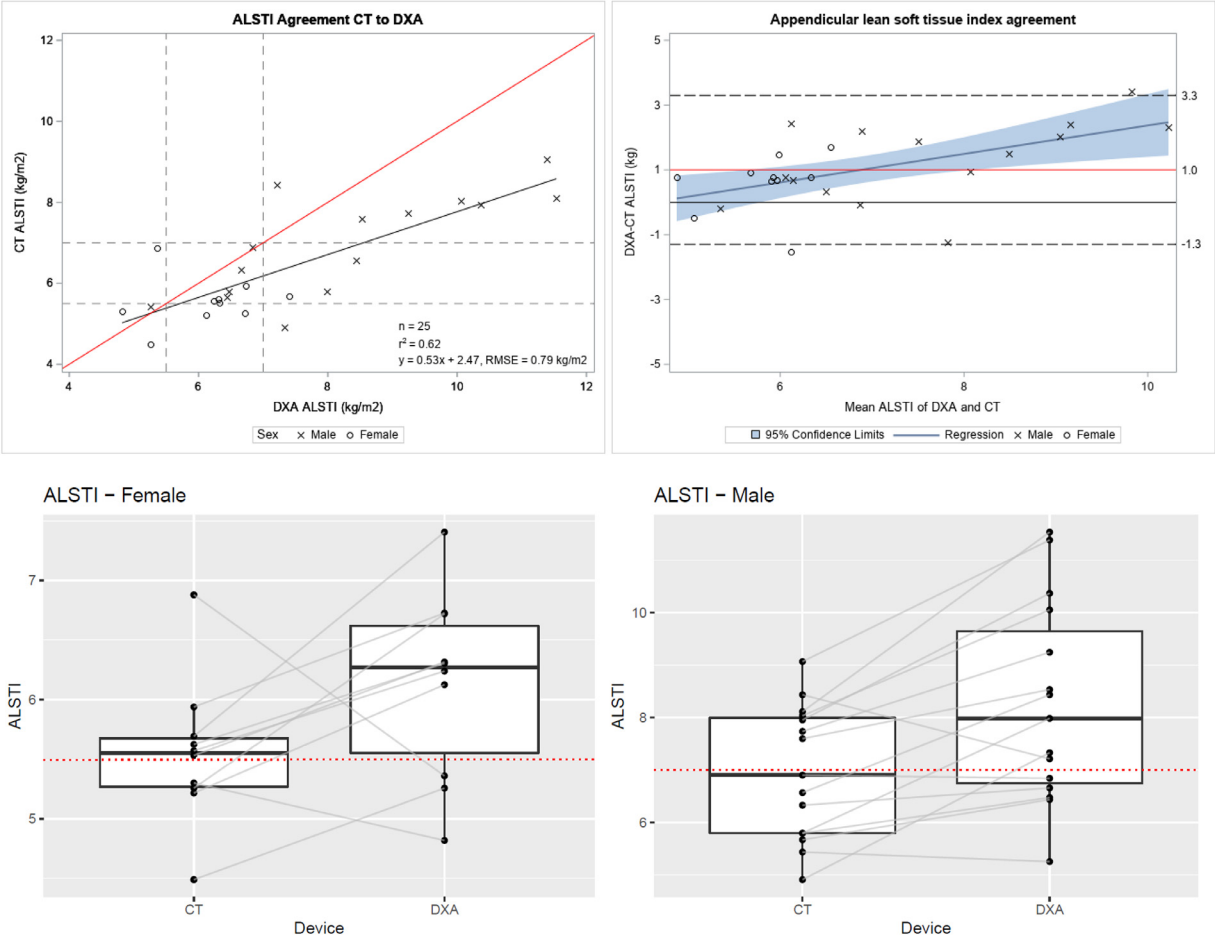

Fig. 2. Comparison of ALSTI between DXA and CT-prediction (n = 25).

are not solely attributable to FFM_{hyd} [62]. We also show that FFM_{hyd} alone cannot correct the disagreement between DXA and 4C.

Variation in soft tissue composition and the presence of edema will impact attenuation of soft tissue compartments. These effects are frequently reported in DXA systems due to their increased availability in research [63,64]. Abdominal edema is also common in patients with cancer, where water impacts the discrimination of components in CT scans [65]. These system-issues are further compounded in obesity, where the water content of muscle is lower compared to normal-weight adults [66]. Because CT provides a measure of muscle volume, where DXA provides a measure of muscle attenuation, technical differences, solid tumors, and edema can ultimately cause differential measures of either ALST or FFM, particularly when measures of trunk CT are used to predict regional muscle as measured by DXA [65,67,68]. These findings support the disagreement in muscle mass classification between CT and DXA in this study. However, the primarily greater ALST (and FFM) by DXA compared to CT indicates that obesity-alone does not drive this disagreement. Given that CT is routinely performed in the oncology setting generally only measuring selected landmarks, single-slice CT scans for whole-body muscle

Table 3
Comparison of prevalence of low lean (or muscle) mass estimations by CT or DXA using published cutoffs (n = 25)

	Total (n = 25), n (%)	Male (n = 15), n (%)	Female (n = 10), n (%)
ALSTI			
DXA	8 (32.5%)	5 (33.3%)	3 (30.0%)*
CT	12 (48.0%)	8 (53.3%)	4 (40.0%)*
SMI			
CT	14 (56.0%)	8 (53.3%)	6 (60.0%)

*One patient was reported as low lean by DXA but not by CT, two were reported as low lean by CT but not by DXA (total of 3 patients differed in classification of low muscle mass).ALSTI low lean: <7.0 kg/m^2 (m), <5.5 kg/m^2 (f).SMI low muscle:if BMI < 30 kg/m2, <52.3 cm^2/m^2 (m), <38.6 cm^2/m^2 (f).if BMI ≥ 30 kg/m2, <54.3 cm^2/m^2 (m), <46.6 cm^2/m^2 (f).

Table 4
Error matrix of predicted low lean mass using cutoffs based on DXA or CT (skeletal muscle index from CT analysis or predicted appendicular lean soft tissue)

	CT SMI		CT-predicted ALSTI	
DXA ALSTI	Low	Normal	Low	Normal
Low	7	1	7	1
Normal	7	10	5	12
Total	14	11	16	9
Kappa	0.39		0.51	
Total misclassified (%)	8 (32.0%)		6 (24.0%)	

Shaded regions indicate false positive or false negative classification of low lean mass.
ALSTI low lean: <7.0 kg/m^2 (m), <5.5 kg/m^2 (f)
SMI low lean:
if BMI < 30 kg/m^2, <52.3 cm^2/m^2 (m), <38.6 cm^2/m^2 (f)
if BMI ≥ 30 kg/m^2, <54.3 cm^2/m^2 (m), <46.6 cm^2/m^2 (f)

quantification and low muscle mass assessment have become increasingly common practice [69], and have proven to be invaluable to the field. That said, DXA measures are more accessible, including for monitoring of change over time, hence its recommendation for clinical estimation of low muscle mass [58,64]. The results of this study show that while hydration may be an important factor for further study, it remains likely that the differences between modalities are the result of improper agreement between thresholds for low muscle mass or inaccurate prediction equations (due to the sample used to derive the equation or possibly from small samples of patients with higher BMI) [41].

Given that all but two CT scans were conducted prior to DXA, which consistently showed higher ALST (and subsequently ALSTI) readings than by CT estimation, factors other than measurement timing impact the agreement between these two modalities. These differences may result from assumptions in the relationship between trunk and whole-body muscle including disproportional changes in muscle atrophy as a result of reductions in weight-bearing activity, hormonal effects that impact regional wasting, or preferential loss of trunk lean mass during cancer progression (with variations in regional lean [decreased arm lean, increased leg lean] due to concomitant onset of inflammation causing muscle loss [arm] and fluid retention [leg]) [70]. Our results suggest that those with higher BMIs were more likely to be classified as having low muscle by CT rather than DXA, emphasizing the need for additional research in patients with excess body weight.

The accuracy of DXA is highlighted by its minimal yet meaningful contribution to the 4C model's estimation of FFM, with BMC accounting for 4.0% to 6.0% of total BM, compared to the contributions from other techniques. This further supports its use as a clinical measurement technique. DXA estimates of visceral adipose tissue have also been calibrated to CT with population-based reference data available, increasing DXA's clinical utility for identification of abnormal visceral fat associated with cardiometabolic disease risk [71,72]. Conversely, ADP is not commonly used in clinical settings and BIA requires specific considerations due to its unique operational principles and requirements [45,68]. Since BIA FFM and ALST are often calibrated against DXA, further work is required to evaluate BIA's accuracy in comparison with criterion measures in oncology contexts [73,74]. While recommendations have been made to develop BIA prediction equations for FFM that account for varying hydration statuses in different populations based on 4C models, the broader LoAs associated with 2C models emphasize the need for refined BIA-based TBW estimations enhancing the precision of more accessible BIA-based 4C models [75,76].

Because patients with colorectal cancer and low muscle mass have significantly lower survival (hazard ratio 1.83, 95% confidence interval [CI]: 1.57−2.14) compared to those with normal muscle mass, the need for an accurate and accessible body composition technique is clear [77]. Overall, results presented support the accuracy of DXA and its utility for regional muscle assessment, though further work is necessary to understand the agreement between CT and DXA and whether improved ALSTI equations are necessary, considering DXA's advantage for more frequent monitoring. Further research will be instrumental in shaping clinical guidelines for use of these modalities for body composition assessment.

The study's primary limitation, as noted above, is the paucity of data on BIA's use for TBW estimation in the context of cancer. Replacing BIA with the gold-standard, deuterium dilution, for TBW assessment reduces mean errors at the group level (the Lukaski equation reported −0.4 L TBW) but larger individuals errors (95% CI of −3.2 to 2.4L) [78]. That said, multicompartment models utilizing TBW equations are reported to be more reliable for FFM assessment than DXA across various ages in healthy adults [11]. This is because the error in TBW estimation is reduced when integrated into the multicompartment model (e.g., error of −0.4 L * 0.699 = 0.280 kg FM, 95% CI: −2.237 to 1.678 kg). Use of three separate BIA devices will inevitably result in differences in raw impedance measurements, however the use of the raw impedance parameters to predict TBW provide greater confidence compared to a BIA prediction equation that was likely developed in healthy adults [79]. Additionally, FFM_{hyd} levels derived from this study identified a wide range of values among patients, highlighting potential issues in the accuracy of DXA or ADP measurements. These findings warrant further studies assessing a criterion multicompartment model in patients with cancer and deriving more accurate BIA-based empirical equations for TBW estimation. Another limitation in the multicompartment modeling was use of an estimated thoracic gas volume for ADP. The error in thoracic gas volume (standard error of estimate = 0.442 L) is considered minimal (0.6% of the average BV in this sample), with the reported mean difference between estimated and measured thoracic volume (−0.05 L) resulting in a 0.14 kg error in FM [80]; this indicates that the effect of this error appears to be minimal.

While CT remains state-of-the-art and provides detail such as muscle composition, application of the Mourtzakis equation for calculating ALST from CT cross-sectional area has its limitations due to its derivation from a small and specific patient cohort [41]. The context-specific development may make the equation prone to error and therefore warrants further work to understand the accuracy and broad applicability of CT for ALST prediction and low muscle mass identification. Given the small sample of data available in this study, larger studies should examine the relationships between CT and DXA to identify potential covariates that may improve the agreement between modalities. Additionally, while some of the CT scans were performed up to 90 d prior to DXA (average of 60 d), the lower ALST by CT shows that potential muscle loss that may have occurred between CT and DXA scans does not explain the observed discrepancies. Future studies examining the agreement between CT and DXA should aim to minimize the time between measurements, by coordinating DXA data collection closely with scheduled CT scans.

Conclusion

These findings support the use of DXA as a clinical assessment method of body composition in patients with cancer, including for estimation of regional muscle mass. Disagreement in ALST between CT (estimated) and DXA means that results from these modalities cannot be interchanged. Baseline and longitudinal assessment with DXA may provide important insight into disease progression and/or treatment outcomes. These findings underscore the need for additional studies on body composition modeling within clinical populations to improve our understanding of the impact that nutritional status and illness have on body composition and its assessment. Implementing multicompartment models has the potential to advance body composition research in oncology, providing a greater understanding of its effects on health outcomes.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Carla M Prado reports financial support was provided by Alberta Government. Katherine Ford reports a relationship with

Abbott Nutrition that includes: consulting or advisory and paid expert testimony. Maria Cristina Gonzalez reports a relationship with Abbott Nutrition that includes: consulting or advisory. Carla M Prado reports a relationship with Abbott Nutrition that includes: consulting or advisory. Maria Cristina Gonzalez reports a relationship with Nutricia Research BV that includes: consulting or advisory. Maria Cristina Gonzalez reports a relationship with Nestle Brazil that includes: consulting or advisory. Carla M Prado reports a relationship with Nutricia Research BV that includes: consulting or advisory. Crala M Prado reports a relationship with Nestle Health Science that includes: consulting or advisory. Crala M Prado reports a relationship with AMRA medical that includes: consulting or advisory. Carla M Prado reports a relationship with Pfizer that includes: consulting or advisory. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jonathan P. Bennett: Conceptualization, Formal analysis, Methodology, Writing – original draft. **Katherine L. Ford:** Writing – original draft, Investigation, Data curation, Conceptualization. **Mario Siervo:** Writing – review & editing, Methodology, Formal analysis. **Maria Cristina Gonzalez:** Writing – review & editing, Conceptualization. **Henry C. Lukaski:** Methodology, Formal analysis, Writing – review & editing. **Michael B. Sawyer:** Methodology, Investigation, Writing – review & editing. **Marina Mourtzakis:** Writing – review & editing, Methodology. **Nicolaas E.P. Deutz:** Methodology, Writing – review & editing. **John A. Shepherd:** Formal analysis, Writing – review & editing. **Carla M. Prado:** Writing – original draft, Supervision, Funding acquisition, Formal analysis, Conceptualization.

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Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.nut.2024.112494.

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